

Machine Learning for Natural Language Processing

Direct Preference Optimization



UNIVERSITY OF
GOTHENBURG

CHALMERS

Richard Johansson

`richajo@chalmers.se`

Recap of RLHF (Ouyang et al., 2022)

We want to train a LM to maximize a **reward** $r(x, y)$ that models high-level text quality

In **RLHF**, the reward is a **learned model** trained on human preferences

Write a slogan about the values of Chalmers.

We stand for inclusiveness, creativity, and sustainability.

I don't understand your request.

The values of Chalmers are limitless.

The reward model is trained using the **Bradley-Terry** binary preference model

$$P(y_1 \succ y_0) = \sigma(r_\theta(x, y_1) - r_\theta(x, y_0))$$

PPO recap (without token prediction loss)

$$\text{obj}(\pi^{\text{RL}}) = \mathbb{E}_{(x,y) \sim D_{\pi^{\text{RL}}}} [\boldsymbol{r}(x, y) - \beta \log(\pi^{\text{RL}}(y|x) / \pi^{\text{SFT}}(y|x))]$$

Fairly complex to implement: it involves **four** different models (π^{RL} , π^{SFT} , $\boldsymbol{r}(x, y)$, and a **value network** for token-level utility)

Direct Preference Optimization: Your Language Model is Secretly a Reward Model

Rafael Rafailov^{*†}

Archit Sharma^{*†}

Eric Mitchell^{*†}

Stefano Ermon[‡]

Christopher D. Manning[†]

Chelsea Finn[†]

[†]Stanford University [‡]CZ Biohub
{rafailov,architsh,eric.mitchell}@cs.stanford.edu

Abstract

While large-scale unsupervised language models (LMs) learn broad world knowledge and some reasoning skills, achieving precise control of their behavior is difficult due to the completely unsupervised nature of their training. Existing methods for gaining such steerability collect human labels of the relative quality of model generations and fine-tune the unsupervised LM to align with these preferences, often with reinforcement learning from human feedback (RLHF). However, RLHF is a complex and often unstable procedure, first fitting a reward model that reflects the human preferences, and then fine-tuning the large unsupervised LM using reinforcement learning to maximize this estimated reward without drifting too far from the original model. In this paper, we leverage a mapping between reward functions and optimal policies to show that this constrained reward maximization problem can be *optimized exactly* with a single stage of policy training, essentially solving a classification problem on the human preference data. The resulting algorithm, which we call *Direct Preference Optimization* (DPO), is stable, performant, and computationally lightweight, eliminating the need for fitting a reward model, sampling from the LM during fine-tuning, or performing significant hyperparameter tuning. Our experiments show that DPO can fine-tune LMs to align with human preferences as well as or better than existing methods. Notably, fine-tuning with DPO exceeds RLHF's ability to control sentiment of generations and improves response quality in summarization and single-turn dialogue while being substantially simpler to implement and train.

Direct Preference Optimization: key points

Trains on preferences **without reinforcement learning**

Simpler to implement: reward model not mandatory, no value network, no sampling/rollouts

Competitive with PPO in practice

DPO: the objective

Given a training prompt x with outputs y_w (winner) and y_l (loser):

$$\mathcal{L}_{\text{DPO}} = -\log \sigma \left(\beta \log \frac{\pi^{\text{DPO}}(y_w|x)}{\pi^{\text{SFT}}(y_w|x)} - \beta \log \frac{\pi^{\text{DPO}}(y_l|x)}{\pi^{\text{SFT}}(y_l|x)} \right)$$

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- $\pi^{\text{DPO}}(y|x)$ is the probability of y in the trained model
- $\pi^{\text{SFT}}(y|x)$ is the probability of y in the reference SFT model
- β controls how much we deviate from the SFT model
- $-\log \sigma$ is equivalent to `BCEWithLogitsLoss`

Training data for DPO

	instruction	winner	loser
0	Compose a story about a magical garden that on...	Once upon a time, in a realm beyond our waking...	Oh my, that sounds like a fascinating and imag...
1	Write a summary of a political article.	The political article discusses the upcoming p...	I am not capable of writing political summarie...
2	Write a 5-paragraph essay discussing at least ...	Heart disease is a leading cause of death worl...	Alternative medicine refers to different medic...
3	Please provide a detailed list of various type...	There are several software life cycle models t...	Software life cycle models are used to describ...
4	Provide step-by-step instructions on how to re...	Step 1: Research and Identify a Market{n1. Fin...	In order to research and identify a market, po...
5	Please develop a Ruby on Rails application tha...	As an AI language model, I'm unable to actual...	As an AI assistant, I can provide the general ...
6	Over 65 graduate and undergraduate students fr...	Besides the University of Maryland, Old Domini...	The other universities that were represented a...
7	"Mama!" the guard says. We follow his pointing...	The guard at Cerro Bandolista shows his passio...	What aspect of the guard at Cerro Bandolista's...
8	Who were some of the influential figures in th...	Thank you for asking! The Abstract Expressioni...	The Abstract Expressionist movement was marked...
9	What are the most unique features of the flora...	The Galapagos Islands are known for their uniq...	The Galapagos Islands are known for their uniq...
10	How do Hindu beliefs and practices intersect w...	Hinduism, being one of the oldest and most div...	Hinduism has a complex relationship with gende...
11	How has the natural environment surrounding Bo...	Confidence: 90%\n\nThe natural environment sur...	The natural environment of Boland, South Afric...
12	How does addiction manifest in the workplace, ...	Addiction in the workplace can manifest in var...	Alcohol and drug addiction can have serious co...
13	Provide a comprehensive explanation of how emo...	Emotional Intelligence (EI), popularized by ps...	Emotional intelligence (EI) plays a crucial ro...
14	Are there alternative forms of protest that ca...	Yes, in times of crisis, virtual forms of prot...	Yes, there are alternative forms of protest th...
15	Write a 1000-word informative article in a for...	Title: Navigating the Toxic Terrain: Strategie...	Workplace bullying and toxic coworkers can hav...
16	What were some of the key themes explored in t...	The Coen Brothers, Joel and Ethan, are America...	Thank you for your question! The Coen Brothers...
17	Build a website for a fitness club that includ...	To build a modern, sleek website for a fitness...	Do you have any specific skills or qualificat...
18	The Support Vector Machine (SVM) is one of the...	The key idea of the SVM classifier presented l...	The key idea of the SVM classifier presented l...
19	Write a 1500-word report on the long-term impa...	Title: The Long-Term Impacts of Fast Food Cons...	The long-term impacts of fast food consumption...

Source:

<https://huggingface.co/datasets/openbmb/UltraFeedback>

Derivation of DPO

For PPO-style RL, it can be shown theoretically that we can write the reward in terms of the **optimal policy**:

$$r(x, y) = \beta \log \frac{\pi^{\text{optimal}}(y|x)}{\pi^{\text{SFT}}(y|x)} + \beta \log Z(x)$$

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Assuming that the reward is **Bradley-Terry**:

$$P(y_w \succ y_l) = \sigma(r(x, y_w) - r(x, y_l))$$

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Assuming that the reward is **Bradley-Terry**:

$$P(y_w \succ y_l) = \sigma(r(x, y_w) - r(x, y_l))$$

Combining the two equations, we get the DPO objective:

$$P(y_w \succ y_l) = \sigma \left(\beta \log \frac{\pi^{\text{DPO}}(y_w|x)}{\pi^{\text{SFT}}(y_w|x)} - \beta \log \frac{\pi^{\text{DPO}}(y_l|x)}{\pi^{\text{SFT}}(y_l|x)} \right)$$

Unpacking DPO and PPO: Disentangling Best Practices for Learning from Preference Feedback

Hamish Ivison[✳] Yizhong Wang[✳] Jiacheng Liu[✳]
Zequ Wu[✳] Valentina Pyatkin[✳] Nathan Lambert[✳]
Noah A. Smith[✳] Yejin Choi[✳] Hannaneh Hajishirzi[✳]

[✳]Allen Institute for AI [✳]University of Washington
hamishiv@cs.washington.edu

Abstract

Learning from preference feedback has emerged as an essential step for improving the generation quality and performance of modern language models (LMs). Despite its widespread use, the way preference-based learning is applied varies wildly, with differing data, learning algorithms, and evaluations used, making disentangling the impact of each aspect difficult. In this work, we identify four core aspects of preference-based learning: **preference data**, **learning algorithm**, **reward model**, and **policy training prompts**, systematically investigate the impact of these components on downstream model performance, and suggest a recipe for strong learning for preference feedback. Our findings indicate that all aspects are important for performance, with better preference data leading to the largest improvements, followed by the choice of learning algorithm, the use of improved reward models, and finally the use of additional unlabeled prompts for policy training. Notably, PPO outperforms DPO by up to 2.5% in math and 1.2% in general domains. High-quality preference data leads to improvements of up to 8% in instruction following and truthfulness. Despite significant gains of up to 5% in mathematical evaluation when scaling up reward models, we surprisingly observe marginal improvements in other categories.

We publicly release the code used for training¹ and evaluating² our models, along with the models and datasets themselves³.

DPO implementations

- Eric Mitchell's original implementation: <https://github.com/eric-mitchell/direct-preference-optimization>
- HuggingFace TRL: https://huggingface.co/docs/trl/dpo_trainer

DPO Trainer

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All models DPO smol course Chapter 2

Overview

TRL supports the DPO Trainer for training language models from preference data, as described in the paper [Direct Preference Optimization: Your Language Model is Secretly a Reward Model](#) by [Rafael Rafailov](#), Archit Sharma, Eric Mitchell, [Stefano Ermon](#), [Christopher D. Manning](#), [Chelsea Finn](#).

The abstract from the paper is the following:

DPO: takeaways

Trains on preference data **directly**, no RL

Competitive with RL-based training approaches

Used in many LLMs (as of 2025): Qwen, Phi, OLMo, Mixtral...

The pipeline is often **SFT→DPO→RL**

References

- H. Ivison, Y. Wang, J. Liu, Z. Wu, V. Pyatkin, N. Lambert, N. A. Smith, Y. Choi, and H. Hajishirzi. 2024. [Unpacking DPO and PPO: disentangling best practices for learning from preference feedback](#). In *NeurIPS*.
- L. Ouyang, J. Wu, and X. Jiang et al. 2022. [Training language models to follow instructions with human feedback](#). arXiv:2203.02155.
- R. Rafailov, A. Sharma, E. Mitchell, C. D. Manning, S. Ermon, and C. Finn. 2023. [Direct preference optimization: Your language model is secretly a reward model](#). In *NeurIPS*.